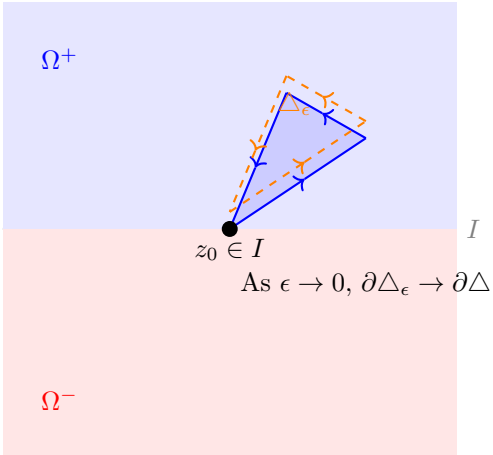
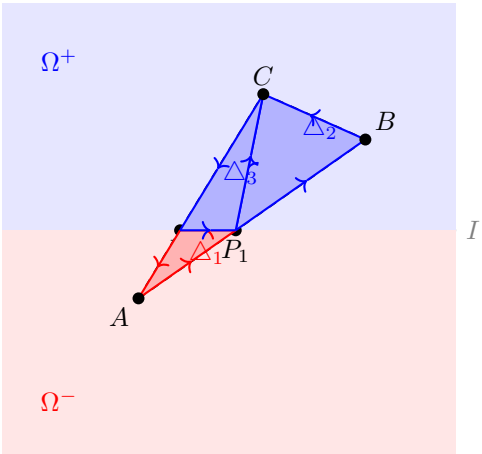


Case 3: If I cuts the triangle then we can split the triangle into smaller triangles and we're done.



The triangle is split into smaller triangles, each either entirely in Ω^+ or Ω^-, or with edges on I, so we can apply Cases 1 and 2. □

10.2 Schwarz Reflection Principle

Theorem 10.4. *Schwarz Reflection Principle: Let $f : \Omega^+ \to \mathbb{C}$ be holomorphic and extend continuous to I with $f^+/I \in \mathbb{R}$ then f^+ extends analytically to Ω*

Proof. Let $f^- : \Omega^- \to \mathbb{C}$ be defined by $f^-(z) = \overline{f^+(\overline{z})}$. We claim f^- is holomorphic on Ω^-, then the previous theorem applies because $f^+ = f^-$ on I.
Fix $z_0 \in \Omega^-$ then $\overline{z_0} \in \Omega^+$, we know $f^+(\overline{z}) = \sum_{n=0}^{\infty} a_n(\overline{z} - \overline{z_0})^n$ for all $\overline{z} \in D(\overline{z_0})$. Thus

$$\overline{f^+(\overline{z})} = \sum_{n=0}^{\infty} \overline{a_n}(\overline{z} - \overline{z_0})^n.$$

We know $\overline{\overline{z} - \overline{z_0}} = z - z_0$ so

$$f^-(z) = \overline{f^+(\overline{z})} = \sum \overline{a_n}(z - z_0)^n.$$

□

11 Feb 9

11.1 Meromorphic Functions and Poles

Definition 11.1. *Meromorphic functions are "determined" by zeros and singularities.*
A point singularity of f is a point $z_0 \in \mathbb{C}$ such that f is defined in a neighborhood of z_0 but not at z_0.

$$\mathcal{D}_\delta(z_0) \setminus \{z_0\}.$$

Example 11.2. *$f(z) = \frac{1}{z}$ has a singularity at $z_0 = 0$.*

Remark 11.3. *Zeros of a holomorphic f are isolated, unless $f \equiv 0$.*

Theorem 11.4. *(Local Description Near Zeros) Let $f : \underset{open}{\Omega} \to \mathbb{C}$ holomorphic and $f(z_0) = 0$, $z_0 \in \Omega, f \not\equiv 0$ then $\exists \mathcal{U} \ni z_0$, $g : \mathcal{U} \to \mathbb{C}$ holomorphic and vanishing ($\forall z \in \mathcal{U}, g(z) \neq 0$) $\exists! n > 0$ such that $f(z) = (z - z_0)^n g(z)$.*

Proof. We know

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

$f \not\equiv 0$ implies that $n =$ smallest k such that $a_k \neq 0$. We can write $f(z) = (z - z_0)^n g(z)$. □

Definition 11.5. *Let $n =$ the multiplicity/order of z_0. When $n = 1$, z_0 is called a simple zero. We say $f : \mathcal{D}_\delta(z_0) \setminus \{z_0\} \to \mathbb{C}$ has a pole at z_0 if $\frac{1}{f}$ extended by 0 at z_0 is holomorphic in $\mathcal{D}_\epsilon(z_0)$, for some $0 < \epsilon < \delta$.*

Theorem 11.6. *(Local Discontinuity near Poles) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ has a pole at z_0 in Ω then $\exists \mathcal{U} \ni z_0$, $\exists h : \mathcal{U} \xrightarrow{holo} \mathbb{C}$ nonvanishing then $\exists! n > 0$ such that $f(z) = (z - z_0)^{-n} h(z)$.*

Proof. Apply previous theorem to $\frac{1}{f}(z) = (z - z_0)^n g(z)$. □

11.2 Laurent Series and Residues

Theorem 11.7. *(Laurent Series Expansion) If f has a pole of order n at z_0 then locally*

$$f(z) = \underbrace{\frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{\overbrace{a_{-1}}^{\text{residue of } f \text{ at } z_0}}{z - z_0}}_{\text{principal part}} + \underbrace{G(z)}_{\text{holo part of } f}.$$

where $G(z)$ is holomorphic and nonvanishing near z_0.
Moreover, $res_{z_0}(f) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}}((z - z_0)^n f(z))$

Proof. By theorem 11.6,

$$f(z) = (z - z_0)^{-n} h(z) = (z - z_0)^{-n} \sum_{k=0}^{\infty} b_k (z - z_0)^k = \frac{b_0}{(z - z_0)^n} + \cdots + \frac{b_{n-1}}{(z - z_0)} + \sum_{k \geq n}^{\infty} b_k (z - z_0)^{k-n}.$$

So we have

$$f(z)(z - z_0)^n = b_0 + b_1(z - z_0) + \cdots + b_{n-1}(z - z_0)^{n-1} + \underbrace{O((z - z_0)^n)}_{\to 0 \text{ as } \frac{d^{n-1}}{dz^{n-1}}}.$$

□

11.3 The Residue Formula

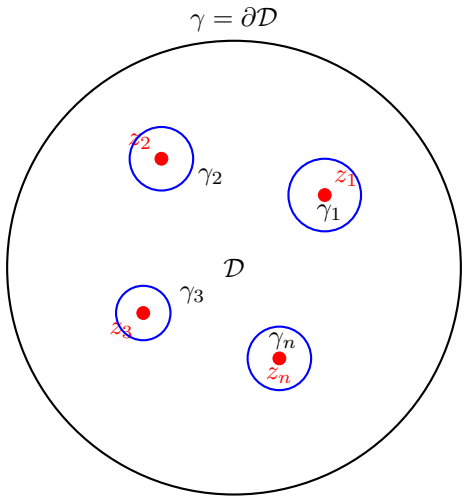
Theorem 11.8. *(The Residue Formula) Suppose* $f : \Omega \setminus \underbrace{\{z_1, z_2, \cdots, z_n\}}_{poles} \to \mathbb{C}$ *is holomorphic. Then for a disk* $\overline{\mathcal{D}} \subset \Omega$ *containing* $z_1, ..., z_n$ *we have*

$$\int_{\partial \mathcal{D}} f = 2\pi i \sum_{k=1}^{n} res_{z_k}(f).$$

Proof. Let $\gamma = \partial \mathcal{D}$ be the boundary of the disk $\mathcal{D}$, oriented counterclockwise. For each pole z_k, choose a small disk $\mathcal{D}_k$ centered at z_k with radius $\varepsilon_k > 0$ small enough so that:

- $\overline{\mathcal{D}_k} \subset \mathcal{D}$ for all k
- $\overline{\mathcal{D}_k} \cap \overline{\mathcal{D}_j} = \emptyset$ for $k \neq j$
- f is holomorphic on $\overline{\mathcal{D}} \setminus \bigcup_{k=1}^{n} \mathcal{D}_k$

Let $\gamma_k = \partial \mathcal{D}_k$ be the boundary of each small disk, oriented clockwise (negative orientation). Consider the multiply connected region $\mathcal{D} \setminus \bigcup_{k=1}^{n} \overline{\mathcal{D}_k}$.



By Cauchy's theorem for multiply connected regions, the integral of f around the boundary of this region (taking into account orientations) is zero. The boundary consists of γ (counterclockwise) and each γ_k (clockwise), so:

$$\int_{\gamma} f(z)dz + \sum_{k=1}^{n} \int_{\gamma_k} f(z)dz = 0.$$

Note that γ_k has clockwise orientation, so reversing it gives:

$$\int_{\gamma} f(z)dz = \sum_{k=1}^{n} \int_{-\gamma_k} f(z)dz.$$

where $-\gamma_k$ denotes γ_k with counterclockwise orientation.
Now, for each pole z_k, consider its Laurent expansion:

$$f(z) = \sum_{j=-\infty}^{\infty} a_j^{(k)}(z - z_k)^j = \frac{a_{-1}^{(k)}}{z - z_k} + \sum_{j \neq -1} a_j^{(k)}(z - z_k)^j.$$

where $a_{-1}^{(k)} = \text{res}_{z_k}(f)$.
For $j \neq -1$, the function $(z - z_k)^j$ has an antiderivative on $\mathbb{C} \setminus \{z_k\}$ (or on all of $\mathbb{C}$ if $j \geq 0$), so by the fundamental theorem of calculus:

$$\int_{-\gamma_k} (z - z_k)^j dz = 0 \quad \text{for } j \neq -1.$$

For $j = -1$, we compute directly. Parameterize $-\gamma_k$ by $z(t) = z_k + \varepsilon_k e^{it}$ for $t \in [0, 2\pi]$:

$$\int_{-\gamma_k} \frac{1}{z - z_k} dz = \int_0^{2\pi} \frac{1}{\varepsilon_k e^{it}} \cdot i\varepsilon_k e^{it} dt = \int_0^{2\pi} i dt = 2\pi i.$$

Therefore,

$$\int_{-\gamma_k} f(z)dz = a_{-1}^{(k)} \cdot 2\pi i = 2\pi i \cdot \mathrm{res}_{z_k}(f).$$

Combining all terms:

$$\int_{\partial\mathcal{D}} f(z)dz = \sum_{k=1}^{n} \int_{-\gamma_k} f(z)dz = 2\pi i \sum_{k=1}^{n} \mathrm{res}_{z_k}(f).$$

$\square$

12 Feb 11

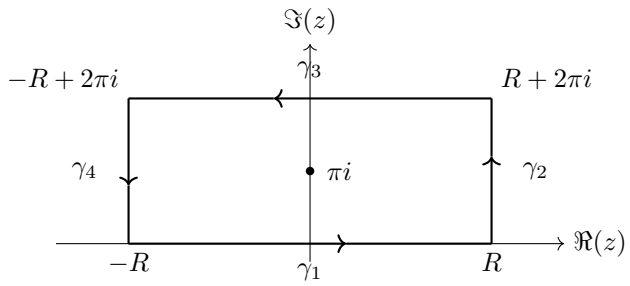
12.1 Integral Evaluation via Residues

Example 12.1.

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = \frac{\pi}{\sin(a\pi)}.$$

Proof. Let $f(z) = \frac{e^{az}}{1+e^z}$ then we have simple poles at $z : e^z = 1$ so $z \in \{..., -\pi i, \pi i, 3\pi i, ...\}$. We have the residual for

$$\int_{\gamma_1} f + \cdots = \int_{\gamma_R} f(z)\, dz = 2\pi i \operatorname{res}_{\pi i}(f).$$



We have

$$\int_{\gamma_1} f(z)dz \xrightarrow{R\to+\infty} \int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = I.$$

On γ_3, $z = x + 2\pi i$ with x from R to $-R$, so

$$\int_{\gamma_3} f(z)\, dz = -\int_{-R}^{R} \frac{e^{a(x+2\pi i)}}{1+e^x} dx = -e^{2\pi a i} \cdot I.$$

As $R \to \infty$, $\int_{\gamma_2} f \to 0$ and $\int_{\gamma_4} f \to 0$. Thus

$$(1 - e^{2\pi a i})I = 2\pi i \cdot \operatorname{res}_{\pi i}(f) = 2\pi i \cdot (-e^{a\pi i}) = -2\pi i e^{a\pi i},$$

$$\left|\int_{\gamma_2} f(z)dz\right| = \left|\int_0^{2\pi} \frac{e^{a(R+iy)}}{1+e^{R+iy}} i dy\right| \leq \int_0^{2\pi} \frac{e^{aR}}{e^R+1} dy \xrightarrow{R\to\infty} 0$$

Additionally $\int_{\gamma_4} f \to 0$. To compute the residue at πi, we have

$$\lim_{z\to\pi i} f(z)(z-\pi i) = \lim_{z\to i\pi} \frac{e^{az}}{\frac{e^z - e^{\pi i}}{z-\pi i}} = \frac{e^{\pi i a}}{e^{\pi i}}.$$

Letting $R \to +\infty$ in the residual formula gives

$$I + 0 + 0 - e^{2\pi a i} I = -2\pi i e^{a\pi i}.$$

So

$$I = -\frac{2\pi i e^{\pi i a}}{1 - e^{2\pi a i}} = \frac{\pi}{\sin(a\pi)}.$$

□

12.2 Removable Singularities

Definition 12.2. *Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ be holomorphic, if we can extend f analytically to z_0 we say that f has a removable singularity at z_0.*

Theorem 12.3. *(Riemann's Theorem) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ be holomorphic, if f is bounded near z_0 then z_0 is a removable singularity.*

Proof. Let $\overline{D} \subset \Omega$ be a disc centered at z_0. We want to use Cauchy's formula

$$\frac{1}{2\pi i}\int_{\partial D}\frac{f(w)}{w-z}dw = g(z), \text{ for all} z \in D.$$

It suffices to show that $f = g$ on $D \setminus \{z_0\}$ because then g is the desired extension of f.
$F(w, z)$ is jointly continuous on $\partial D \times \overline{D}_{r-\epsilon}(z_0)$ and $\forall w \in \partial D$, $z \mapsto F(w, z)$ is holomorphic on D.
Fix $z \in D \setminus \{z_0\}$ then

$$\int_\gamma \frac{f(w)}{w-z}dw = 0 \implies \int_{\partial D}\frac{f(w)}{w-z}dw = \int_{C_\epsilon(z_0)}\frac{f(w)}{w-z}dw + \int_{C_\epsilon(z)}\frac{f(w)}{w-z}dw = 0 + 2\pi i f(z).$$

□

13 Feb 13

13.1 Classification of Singularities

Theorem 13.1. *(Riemann) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ has a removable singularity at z_0 if and only if bounded near z_0.*

Corollary 13.2. *Pole at z_0 if and only if $|f| \to +\infty$ as $z \to z_0$*

Proof. ($\Longrightarrow$) Local description near poles implies $f(z) = (z - z_0)^{-n} g(z)$ so $|f| \to +\infty$
($\Longleftarrow$) Since $\left|\frac{1}{f}\right| \to 0$, in particular $\frac{1}{f}$ is bounded near z_0, Riemann implies $\frac{1}{f}$ has a removable singularity at z_0. □

Theorem 13.3. *(Casorati-Weierstrass) If $f : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}$ is holomorphic with an essential singularity at z_0 then the image of f is dense in $\mathbb{C}$.*

Proof. If not, $\exists w \in \mathbb{C}$ such that no values of f near w, $\exists \delta > 0$ s.t. $\forall z$, $|f(z) - w| > \delta$.
Consider

$$g(z) = \frac{1}{f(z) - w} : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}.$$

Such a g is bounded by $\frac{1}{\delta}$. So g extends halomorphically into $D_r(z_0)$.

(Case 1) $g(z_0) = 0 \implies \frac{1}{g} \to +\infty$ so $f(z) - w$ has a pole so f has a pole

(Case 2) $g(z_0) \neq 0 \implies \frac{1}{g}$ is well defined and f has a removable singularity.

□

Theorem 13.4. *(Picard's Theorem) If $f : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}$ is holomorphic with an essential singularity at z_0 then the image of f is either $\mathbb{C}$ or $\mathbb{C}$ except for at most one value.*

13.2 Meromorphic Functions and Behavior at Infinity

Definition 13.5. *Let $\Omega \subset \mathbb{C}$ be open. A function f is called meromorphic on Ω if $\exists z_1, z_2, \ldots$ without a limit point such that f is holomorphic in $\Omega \setminus \{z_1, z_2, \ldots\}$ and has poles at $z_1, z_2, \ldots$.*

Theorem 13.6. *(Behavior at $+\infty$) Suppose f is holomorphic in a neighborhood of $_\infty$ i.e. $\{|z| > R\}$ for some $R > 0$.*
Note: $F(z) = f(\frac{1}{z})$ is holomorphic in $D_{\frac{1}{R}}(0) \setminus \{0\}$.
We say f has a removable singularity at ∞ if F has a removable singularity at $z = 0$.

Example 13.7. *e^z has an essential singularity at $+\infty$.*

Remark 13.8. *We'll denote $\overline{\mathbb{C}} = \mathbb{C} \cup \{+\infty\}$*

Example 13.9. *Examples of meromorphic functions on $\mathbb{C}$*

1. *Rational functions: $\frac{P}{Q}$*
2. *The gamma function: $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$ (simple poles at $\ldots, -2, -1, 0$)*
3. *The Riemann zeta function: $\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$ (simple poles at $s = -1, -2, -3, \ldots$)*

14 Feb 16

14.1 Rational Functions via Meromorphy

Theorem 14.1. *Suppose f is meromorphic on $\mathbb{C}$ with a removable singularity or a pole at $+\infty$ then f is a rational function.*

Proof. Claim: Under these assumptions, f has finitely many poles.
If not, $z_1, z_2, ...$ are poles and this sequence cannot be bounded otherwise it got a limit point. Say $|z_k| \to +\infty$, the neighborhoods of ∞ are $\{|z| > R\}$ so f not holomorphic for any $R > 0$ which contradicts that f has a removable singularity at $+\infty$.

For a function F which has a pole at $z = 0$ we have

$$F(z) = \underbrace{\frac{a_n}{z^n} + ... + \frac{a_{-1}}{z}}_{f(z)} + \underbrace{a_0 + a_1 z + \cdots}_{g(z)}.$$

Let $z_1, \ldots, z_N$ be the poles of f. Near z_k, for each $1 \leq k \leq N$,

$$f(z) = \underbrace{f_k(z)}_{\substack{\text{principal part} \\ \text{poly in } \frac{1}{z-z_k}}} + \underbrace{g_k(z)}_{\text{holomorphic}}.$$

Near ∞,

$$f\left(\frac{1}{z}\right) = \underbrace{\tilde{f}_\infty(z)}_{\substack{\text{principal part} \\ \text{poly in } \frac{1}{z}}} + \underbrace{\tilde{g}_\infty(z)}_{\text{holomorphic}}.$$

Consider,

$$H(z) = f(z) - \underbrace{\sum_{j=1}^{N} f_j(z)}_{\text{rational}} - \underbrace{\tilde{f}_\infty(z)}_{\text{poly}}.$$

Claim: H is holomorphic and bounded.
Near z_k

$$H = \underbrace{f - f_k}_{\text{holo}} - \sum_{j \neq k} f_j - \underbrace{f_\infty}_{\text{holo}}.$$

We can show H is bounded as

$$H\left(\frac{1}{z}\right) = \underbrace{f\left(\frac{1}{z}\right) - \tilde{f}_\infty(z)}_{\substack{\tilde{g}_\infty(z) \\ \text{bounded near } z=0}} - \underbrace{\sum_{j=1}^{N} f_j\left(\frac{1}{z}\right)}_{\substack{\text{poly} \\ \text{bounded near } z=0}}.$$

So $H(\frac{1}{z})$ is bounded in say $\{|z| < \delta\}$ and $H(z)$ is bounded in $\{|z| > 1/\delta\}$, hence H is bounded in $\{|z| \leq 1/\delta\}$. *dset* □

15 Feb 18

15.1 Argument Principle and Rouche's Theorem

Theorem 15.1. *(The Argument Principle) Let $f : \Omega \to \mathbb{C}$ be meromorphic, $\overline{D} \subset \Omega$ be a disk. If If f never vanishes on ∂D nor has nay poles on ∂D then*

$$\frac{1}{2\pi i}\int_{\partial D}\frac{f'}{f} = \#\text{ zeros of } f - \#\text{ poles of } f \text{ in } \overline{D}.$$

Proof. If f has zero of order n at z_0 then $f(z) = (z - z_0)^n g(z)$ nonvanishing near z_0 then

$$\frac{f'}{f}(z) = \frac{n}{z - z_0} + \frac{g'}{g}(z).$$

If f has a pole of order n at z_0 we can write $f(z) = (z - z_0)^{-n} h(z)$ then

$$\frac{f'}{f}(z) = -\frac{n}{z - z_0} + \frac{h'}{h}(z).$$

Therefore $\frac{f'}{f}$ has simple poles at zeros and poles of f with residue $\frac{\text{order}}{-\text{order}}$. The residue formula finishes the proof. □

Theorem 15.2. *Rouche's Theorem: Let f, g be holomorphic in $\Omega \supset \overline{D}$ with Ω open. If $|f| > |g|$ on ∂D then $f, f + g$ have the same number of zeros in D.*

Proof. Let $f_t = f + t \cdot g$ for $t \in [0, 1]$ and $n_t = \#$ of zeros of f_t in D. By argument principle we have

$$n_t = \frac{1}{2\pi i}\int_{\partial D}\frac{f_t'}{f_t}.$$

f_t never vanishes on ∂D. We can conclude

$$|f_t| \geq |f| - t|g| \geq |f| - |g| > 0.$$

We can show continuity of the integrand by

$$\int_0^{2\pi} \underbrace{\frac{f_t'(z_0 + re^{i\theta})}{f_t(z_0 + re^{i\theta})} rie^{i\theta}}_{H(t,\theta)}\, d\theta.$$

Then $H(t, \theta)$ is jointly continuous in (t, θ). So n_t is continuous and discrete so n_t is constant. □

Proof. Third Proof of fundamental Theorem of Algebra.

$$P(z) = \underbrace{a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}}_{g(z)} + \underbrace{a_n z^n}_{f(z)}.$$

There exist $\mathbb{R} > 0$ such that $|g(z)| < |f(z)|$ on $|z| = R$ □

15.2 Open Mapping and Maximum Modulus Theorems

Theorem 15.3. *(The Open Mapping Theorem) Let $f : \Omega \to \mathbb{C}$ holomorphic, f non-constant then f maps open sets to open sets.*

Proof. Let $w_0 = f(z_0) \in f(\Omega)$. We want to show that every w near w_0 is also in $f(\Omega)$ Fix w near w_0 then consider,

$$h(z) = f(z) - w = \underbrace{(f(z) - w_0)}_{F(z)} + \underbrace{(w_0 - w)}_{G(z)}.$$

Let $\delta > 0$ be such that $\overline{D}_\delta(z_0) \subset \Omega$ and $\forall z \in \partial D_\delta(z_0)$, $f(z) \neq w_0$. Otherwise if there exist no such δ then there exist $(z_n) \to z_0$ such that $f(z_n) = w_0$ for all n then by rigidity f is constant which contradicts the assumption that f is non-constant. Let

$$\epsilon = \inf_{\partial D_\delta(z_0)} |f(z) - w_0| > 0.$$

Now $|G| > |F|$ on $\partial D_\delta(z_0)$ as long as $|w - w_0| < \epsilon$. By Rouche's Theorem, $h = F + G$ has a zero in $\overline{D}_\delta(z_0)$. □

Theorem 15.4. *(The Maximum Modulus Theorem) Let $f : \Omega \to \mathbb{C}$ holomorphic. If f is holomorphic and non-constant then $|f|$ cannot attain a maximum in Ω.*

Proof. AFSOC, $\exists z_0 \in \Omega$ such that $|\underbrace{f(z_0)}_{w_0}| = \sup_\Omega |f|$ then this contradicts the open mapping theorem because there exist w near w_0 with $|w| > |w_0|$ and the value w must be obtained. □

Corollary 15.5. *If Ω additionally bounded ($\overline{\Omega}$ compact) then $\sup_\Omega |f| = \max_{\partial\Omega} |f|$*

16 Feb 20

16.1 Homotopy and Path Independence

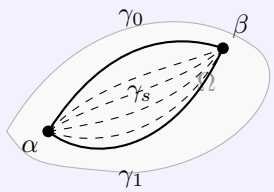
Definition 16.1. *Let $\Omega \subset \mathbb{C}$ be a region, two curves in Ω say with parametrization $\gamma_0, \gamma_1 : [a,b] \to \mathbb{C}$ and common endpoints*

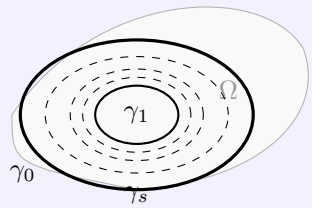
$$\alpha = \gamma_0(a) = \gamma_1(a) \text{ and } \beta = \gamma_0(b) = \gamma_1(b).$$

are called homotopic if $\forall s \in [0,1]$, $\exists \gamma_s : [a,b] \to \Omega$ such that

1. *$\gamma_{s=0} = \gamma_0$ and $\gamma_{s=1} = \gamma_1$,*
2. *$\gamma_s(a) = \alpha$ and $\gamma_s(b) = \beta$ for all $s \in [0,1]$,*
3. *the map $(s,t) \mapsto \gamma_s(t)$ is continuous on $[0,1] \times [a,b]$.*

The map $H(s,t) = \gamma_s(t)$ is called a homotopy between γ_0 and γ_1.





Theorem 16.2. *If $f : \Omega \to \mathbb{C}$ be holomorphic, γ_0, γ_1 are two homotopic curves in Ω then $\int_{\gamma_0} f = \int_{\gamma_1} f$*

Proof. Let $H(s,t) : [0,1] \times [a,b] \to \Omega$ be a homotopy between γ_0, γ_1. Since $K = H([0,1] \times [a,b])$ is compact in open Ω, $\exists \epsilon > 0$ such that $\forall z \in K$, $D_{3\epsilon}(z) \subset \Omega$. Fix such $\epsilon > 0$ and take $\delta > 0$ such that

$$|H(s_1,t_1) - H(s_2,t_2)| < \epsilon \text{ provided that } |(s_1,t_1) - (s_2,t_2)| < 2\delta. \qquad \text{(unif cont of } H\text{)}$$

Given such δ, it suffices to show

$$\int_{\gamma_s} f = \int_{\gamma_{s'}} f \text{ for all } 0 \le s < s' \le 1 \text{ and } |s - s'| < \delta.$$

Take $a = t_0 < t_1 < \cdots < t_n < t_{n+1} = b$ such that $|t_{i+1} - t_i| < \delta$ for all $i = 0, 1, \cdots, n$.
Let's denote $z_i = \gamma_s(t_i)$ and $w_i = \gamma_{s'}(t_i)$. Let $D_i = D_{z_i}(2\epsilon)$.

Claim 16.3.

1. *$z_i, z_{i+1} \in D_i$ holds because $|z_{i+1} - z_i| < \epsilon$ as $z_{i+1} = \gamma_s(t_{i+1}) = H(s, t_{i+1})$ and $z_i = \gamma_s(t_i) = H(s, t_i)$.*
2. *$w_i, w_{i+1} \in D_i$ holds because $|w_i - z_i| < \epsilon$ so*
$$|w_{i+1} - z_i| \le |w_{i+1} - z_{i+1}| + |z_{i+1} - z_i| < \epsilon + \epsilon = 2\epsilon.$$
3. *$D_0 \cup D_1 \cup \cdots \cup D_n$ covers $\gamma_s, \gamma_{s'}$ because for $z = \gamma_s(t)$, $t_i \le t \le t_{i+1}$, $|z - z_i| = |\gamma_s(t) - \gamma_s(t_i)| \le \epsilon$ so $z \in D_i$. For $w = \gamma_{s'}(t)$, $|w - z_i| = |\gamma_{s'}(t) - \gamma_s(t_i)| < \epsilon$.*

Finally, let F_i be a primitive of f on D_i (exists by Cauchy applied to D_i). On $D_i \cap D_{i+1}$, F_i, F_{i+1} are two primitives of the same function f so

$$F_{i+1}(w) - F_{i+1}(z) = c_i.$$

where c_i is a constant. Then

$$\gamma_{\gamma_s} f = \int_a^b f(\gamma_s(t))\gamma_s'(t)dt = \sum_{i=0}^{n} \int_{t_i}^{t_{i+1}} f(\gamma_s(t))\gamma_s'(t)dt = \sum_{i=0}^{n} F_i(z_{i+1}) - F_i(z_i).$$

So

$$\begin{aligned}
\int_{\gamma_s} f - \int_{\gamma_{s'}} f &= \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n} [F_{i+1}(w_{i+1}) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n} [F_i(z_i) - F_{i+1}(w_i)] \\
&= F_0(w_0) - F_0(z_0) + \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n-1} [F_i(z_i) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n-1} [F_i(z_i) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(z_i) + F_{i+1}(z_i) - F_i(w_{i+1})] \\
&= \sum_{i=0}^{n-1} -c_i + c_i = 0
\end{aligned}$$

□

16.2 Simply Connected Regions

Definition 16.4. *A region $\Omega \subset \mathbb{C}$ is simply connected if every two curves in Ω with common endpoints are homotopic.*

Example 16.5. *Examples of simply connected regions:*

1. *Convex Sets*
2. $\Omega = \mathbb{C} \setminus (-\infty, 0]$

Theorem 16.6. *For all f holomorphic in a simply connected region then for closed γ*

$$\int_\gamma f = 0.$$

Theorem 16.7. *f holomorphic has a primitive in a simply connected region Ω.*

17 Feb 23

17.1 Complex Logarithm

Theorem 17.1. *(Construction of Complex Logarithm) Let Ω be a simply connected region such that $0 \notin \Omega$ but $1 \in \Omega$. Then there $\exists!$ holomorphic F on Ω with $F(z) = \log \Omega$ such that*

1. $e^{F(z)} = z$ *on* Ω
2. $F(r) = \log r \ \forall r \in \Omega \cap (1-\delta, 1+\delta)$

Proof. Let $F(z) = \int_{\gamma_z} \frac{dw}{w}$ where γ_z is a curve $1 \curvearrowright z$ in Ω. This is well defined as it doesn't depend on choice of γ_z because Ω is simply connected.
Then we know $F'(z) = \frac{1}{z}$.
Thus

1. $(ze^{-F(z)})' = e^{-F(z)} - ze^{-F(z)}\frac{1}{z} = 0$ so $ze^{-F(z)} = 1$ at $z = 1$.
2. $F(r) = \int_1^r \frac{dt}{t} = \log t \,|_{t=1}^{t=r} = \log r$

□

Example 17.2. $\Omega_0 = \mathbb{C} \setminus (-\infty, 0]$

Claim 17.3. *For $z = re^{i\theta}$ for $|\theta| < \pi$, $\log(z) = \log r + i\theta$, we take the path $1 \to r \to z$. So*

$$\log z = \int_{\gamma_z} \frac{dw}{w} = \int_1^r \frac{dt}{t} + \int_0^{2\pi} \frac{rie^{it}}{re^{it}} dt = \log r + i\theta.$$

Remark 17.4. $\log(z_1 z_2) \neq \log z_1 + \log z_2$ *is in general false.*

Theorem 17.5. *For the principle branch*

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots. \qquad \text{(For } |z| < 1\text{)}$$

Proof. $\frac{d}{dz}(\text{LHS} - \text{RHS}) = \frac{1}{1+z} - \frac{1}{1+z} = 0$ □

17.2 General Powers and Branches

Definition 17.6. *Given $\alpha \in \mathbb{C}$ we define*

$$z^\alpha := e^{\alpha \log_\Omega z}.$$

Example 17.7. *We have*

1. $1^\alpha = 1$
2. $\left(z^{\frac{1}{n}}\right)^n = z^{\frac{1}{n}} \cdot z^{\frac{1}{n}} \cdots z^{\frac{1}{n}} = \left(e^{\frac{1}{n}\log z}\right)^n = e^{\log z} = z$

Theorem 17.8. *Given f holomorphic, nowhere vanishing there exist g such that $e^g = f$*

Proof. Repeat the previous $g' = \frac{f'}{f}$ with $g(z) = \int_{\gamma_z} \frac{f'(w)}{f(w)} dw$

□

18 Feb 25

Definition 18.1. *For $f \in L_1$ $(\mathbb{R} \to \mathbb{C})$ i.e. $||f||_{L_1} = \int_{-\infty}^{+\infty} |f| < \infty$ we define its fourier transform*

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i t \zeta} dt. \qquad (\zeta \in \mathbb{R})$$

Example 18.2. *Some examples of fourier transforms:*

1. $\widehat{e^{-\pi x^2}}(\zeta) = e^{-\pi \zeta^2}$
2. $\widehat{\frac{1}{\cosh(\pi x)}}(\zeta) = \frac{1}{\cosh(\pi \zeta)}$
3. $\widehat{1_{[-1,1]}}(\zeta) = \int_{-1}^{1} e^{-2\pi i \zeta t} dt$

Lemma 18.3. *Basic Properties of Fourier Transforms:*

1. *$\hat{f}$ is uniformly continuous*
2. *$||\hat{f}||_\infty \leq ||f||_{L_1}$*
3. *$\hat{f}(\zeta) \to 0$ as $|\zeta| \to \infty$*

Theorem 18.4. *(Paley-Wiener Theorem) Regularity of f $\iff$ decay of f^**

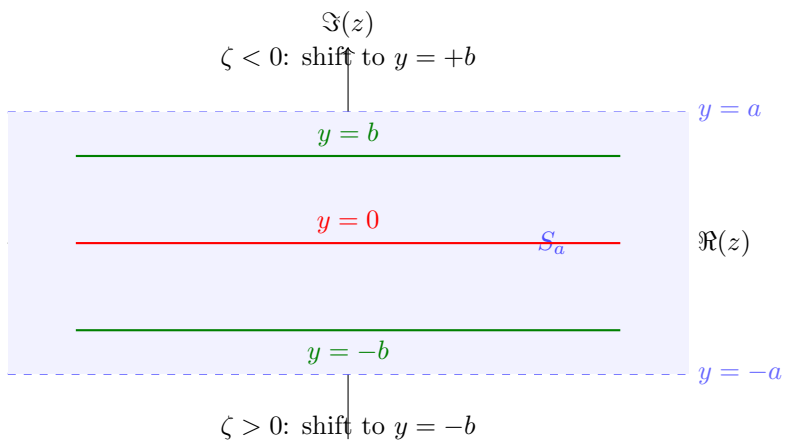
Definition 18.5. *Given $a > 0$, we let $S_a = \{z \in \mathbb{C} : |Im(z)| < a\}$ and*

$$\mathcal{F}_a = \left\{ f : S_a \to \mathbb{C},\ holo\ , \exists C_f, \forall x + iy \in S_a, |f(x = iy)| \leq \frac{C_f}{1 + x^2} \right\}.$$

Additionally, $\mathcal{F} = \bigcup_{a>0} \mathcal{F}_a$

Theorem 18.6. *If $f \in \mathcal{F}_a$, then $|\hat{f}(\zeta)| \leq C e^{-2\pi b |\zeta|}$, for $\zeta \in \mathbb{R}$ and $\forall b < a$.*

Proof. Fix b with $0 < b < a$. The idea is to deform the contour of integration in the Fourier integral so that the factor $e^{-2\pi i \zeta z}$ gains exponential decay. Write $z = x + iy$.



Case $\zeta > 0$: Deform the contour from $\mathbb{R}$ to the line $\{z : \Im(z) = -b\}$. For $z = x - ib$, we have $e^{-2\pi i \zeta z} = e^{-2\pi i \zeta x} \cdot e^{-2\pi \zeta b}$, so the integrand gains a factor $e^{-2\pi \zeta b}$. By Cauchy's theorem (the vertical segments at $\pm\infty$ contribute nothing since $|f(x + iy)| \leq C_f/(1 + x^2)$),

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x - ib)\, e^{-2\pi i \zeta (x - ib)}\, dx = e^{-2\pi \zeta b} \int_{-\infty}^{\infty} f(x - ib)\, e^{-2\pi i \zeta x}\, dx.$$

Hence $|\hat{f}(\zeta)| \leq e^{-2\pi\zeta b} \int_{-\infty}^{\infty} \frac{C_f}{1+x^2}\, dx = C_f \pi\, e^{-2\pi\zeta b} = C\, e^{-2\pi b|\zeta|}$.

Case $\zeta < 0$: Deform to $\{z : \Im(z) = +b\}$. For $z = x + ib$, we have $e^{-2\pi i\zeta z} = e^{-2\pi i\zeta x} \cdot e^{2\pi\zeta b}$ (since $\zeta < 0$, $e^{2\pi\zeta b} = e^{-2\pi|\zeta|b}$). The same argument gives $|\hat{f}(\zeta)| \leq C\, e^{-2\pi b|\zeta|}$. $\square$